

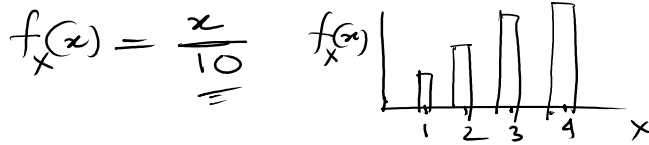
$$P(X=1) = P(X=1, Y=1) + P(X=1, Y=2) \\ + P(X=1, Y=3) + P(X=1, Y=4)$$

12

$$f_{X,Y} = \frac{x}{35 \times 2^{y-2}}, \quad X \in \{1, 2, 3, 4\}, \quad Y \in \{1, 2, 3\}$$

① Marginal distribution of X .

$$f_X(x) = P(X=x) = \sum_{y=1,2,3} \frac{x}{35 \times 2^{y-2}} = \frac{x}{35} \left(\frac{1}{2^{-1}} + \frac{1}{2^0} + \frac{1}{2^1} \right) = \frac{x}{35} \cdot 7 = \frac{x}{5}$$



$$\frac{1}{10} + \frac{2}{10} + \frac{3}{10} + \frac{4}{10} = 1$$

② Marginal distribution of Y

$$f_Y(y) = P(Y=y) = \sum_x \frac{x}{35 \times 2^{y-2}} = \frac{1}{2^{y-2}} \left(\frac{10}{35} \right) = \frac{2}{2^{y-2} \cdot 7} = \frac{8}{2^y \cdot 7}$$

$$f_Y(1) = \frac{8}{14}, \quad f_Y(2) = \frac{8}{28}, \quad f_Y(3) = \frac{1}{7}$$

$$\sum f_Y(y) = \frac{16+8+4}{28} = \frac{28}{28} = 1$$

③ Conditional distribution.

$$\begin{aligned} f_{X|Y}(x,y) &= \mathbb{P}(X=x | Y=y) = \frac{\mathbb{P}(X=x, Y=y)}{\mathbb{P}(Y=y)} \\ &= \frac{\frac{x}{35 \cdot 2^{y-2}}}{\frac{10}{35 \cdot 2^{y-2}}} = \boxed{\frac{x}{10}} \leftarrow \end{aligned}$$

$f_X(x) = f_{X|Y}(x,y)$, X, Y are independent.

④ $f_{Y|X}(x,y) = \mathbb{P}(Y=y | X=x)$

$$P(X=2) = 0$$

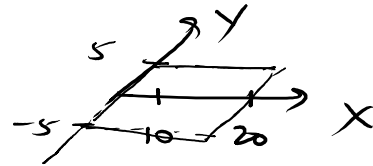
$$f_{X,Y}(x,y) = \frac{2x+y}{C}$$

$$10 \leq X \leq 20$$

$$-5 \leq Y \leq 5$$

① $C = ?$

$$\int_{-5}^5 \int_{10}^{20} \frac{2x+y}{C} dx dy = 1$$



$$\begin{aligned} \frac{1}{C} \int_{-5}^5 [x^2 + yx]_{10}^{20} dy &= \frac{1}{C} \int_{-5}^5 (300 + 10y) dy \\ &= \frac{1}{C} [300y + 5y^2]_{-5}^5 = \frac{1}{C} [3000] \end{aligned}$$

$$C = 3000$$

$$f_{X,Y}(x,y) = \frac{2x+y}{3000}$$

$$(2) \mathbb{P}(10 \leq X \leq 15, 0 \leq Y \leq 5)$$

$$= \int_0^5 \int_{10}^{15} \frac{2x+y}{3000} dx dy = \frac{1375}{6000}$$

$$(3) \text{ Marginal density of } X := f_X(x)$$

$$f_X(x) = \int_{-5}^5 \frac{2x+y}{3000} dy$$

$$= \frac{1}{3000} \left[2xy + \frac{y^2}{2} \right]_{-5}^5 = \frac{20x}{3000} = \frac{x}{150}$$

$$f_X(x) = \frac{x}{150}$$

$$\mathbb{P}(12 \leq X \leq 14) = \int_{12}^{14} \frac{x}{150} dx = \frac{1}{150} \cdot \left(\frac{14^2 - 12^2}{2} \right) = \frac{52}{300}$$

$$(a) f_Y(y) = ?$$

$$f_Y(y) = \int_{10}^{20} \frac{2x+y}{3000} dx$$

$$= \frac{1}{3000} \left[x^2 + yx \right]_{10}^{20} = \frac{1}{3000} (300 + 10y) = \frac{y+30}{300}$$

$$P[0 \leq Y \leq 5] = \int_0^5 \frac{y+30}{300} dy = \frac{1}{2} \text{ (HW)}$$

$$\begin{aligned}
 (5) \quad \underline{\underline{f_{x|y}}}(x,y) &= \mathbb{P}(X=x | Y=y) = \frac{\underline{\underline{f_{x,y}}}(x,y)}{f_Y(y)} \\
 &= \frac{\frac{2x+y}{3000}}{\frac{10y+300}{3000}} = \frac{2x+\textcircled{y}}{10\textcircled{y}+300} \neq \frac{x}{150}
 \end{aligned}$$

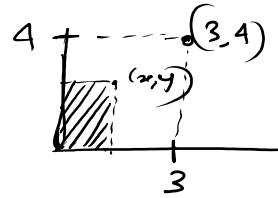
X, Y are not independent

$$F_{x,y}(x,y) = c(\underline{\underline{xy + xy^2}}) \quad , \quad \underline{0 \leq x \leq 3}, \quad \underline{0 \leq y \leq 4}.$$

$$\textcircled{1} \quad F_{x,y}(3,4) = 1$$

$$c(36 + 48) = 1$$

$$\Rightarrow c = 1/84.$$



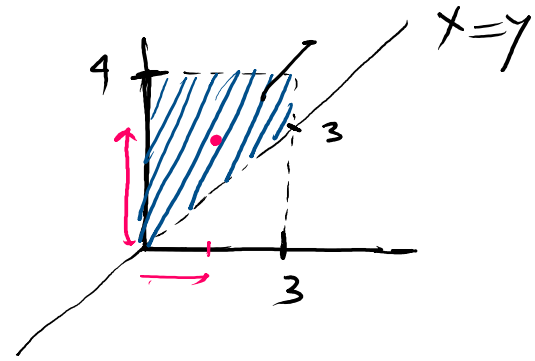
$$\textcircled{2} \quad f_{x,y}(x,y) = \frac{\partial^2 F}{\partial x \partial y} = \frac{1}{84} \frac{\partial}{\partial y} (2xy + y^2) = \frac{1}{84} (2x + 2y) = \underline{\underline{\left(\frac{x+y}{42} \right)}}$$

$$\underline{\underline{P(X \leq Y)}}, \quad \underline{\underline{P(X \geq Y)}}$$

$$= \frac{1}{42} \int_0^3 \int_x^4 (x+y) dy dx$$

$$\textcircled{4w} \rightarrow \underline{\underline{\left(1 - \frac{27}{84}\right)}}$$

$$P(X \geq Y) = \textcircled{\frac{27}{84}}$$



$$= \textcircled{\frac{57}{84}}$$

$$\begin{aligned}
 \text{Cov}(X, Y) &= \mathbb{E}[(X - \mathbb{E}(X))(Y - \mathbb{E}(Y))] \\
 &= \mathbb{E}[XY - X\mathbb{E}(Y) - Y\mathbb{E}(X) + \mathbb{E}(X)\mathbb{E}(Y)] \\
 &= \mathbb{E}[XY] - \mathbb{E}(Y)\mathbb{E}(X) - \mathbb{E}(X)\mathbb{E}(Y) + \mathbb{E}(X)\mathbb{E}(Y) \\
 &= \mathbb{E}[XY] - \mathbb{E}(X)\mathbb{E}(Y) \rightarrow
 \end{aligned}$$

$$\begin{aligned}
 \text{Var}(X) &= \mathbb{E}[(X - \mathbb{E}(X))^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \\
 &= \text{Cov}(X, X)
 \end{aligned}$$